

Chapter 8

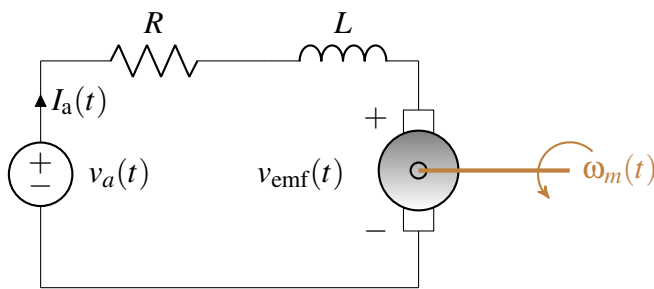
DC Servo Motor

Out of the box a DC Servo motor has a voltage input to the armature, which is $V_a(t)$ and produces an output rotational rate of the motor shaft, namely, $\omega_m(t)$.

Practical applications include using the shaft rotation to:

1. Propel a cart forward by mechanically connecting the shaft to a drivetrain
2. Provide thrust for an aerial vehicle by mechanically connecting the shaft to a fan
3. Produce mechanical energy for conveyor belts

To construct a block diagram and transfer function for the system, we first begin with a circuit diagram for a DC servo



KVL about the circuit in the Laplace domain

$$-V_a(s) + RI_a(s) + LsI_a(s) + V_{\text{emf}}(s) = 0$$

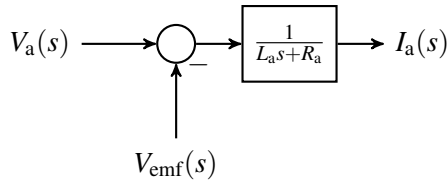
Rewritten so the output is $I_a(s)$

$$(Ls + R)I_a(s) = V_a(s) - V_{\text{emf}}(s)$$

or

$$I_a(s) = \frac{V_a(s) - V_{\text{emf}}(s)}{Ls + R}$$

Let's draw a block diagram for this equation



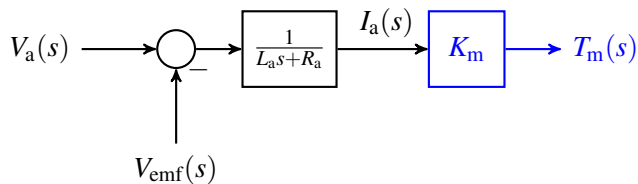
Now, the current $I_a(s)$ through the armature produces a motor torque. The ratio of torque to current is a constant, K_m .

$$K_m = \frac{T_m(s)}{I_a(s)}$$

or

$$T_m(s) = K_m I_a(s)$$

Add this to the equation to the block diagram we are building



For the torque produced in the motor to rotate the rod it must overcome inertia and friction. The inertia is that of the motor itself and the load the motor is driving. Fortunately, the total inertia is simply the summation of the motor and the load

$$J_t = J_m + J_L$$

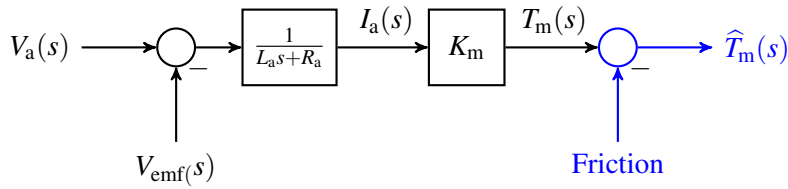
Notes on inertia:

1. For most applications the load inertia is an order of magnitude or greater than the motor inertia.
2. Inertia is the resistance of any physical object to changes in velocity.

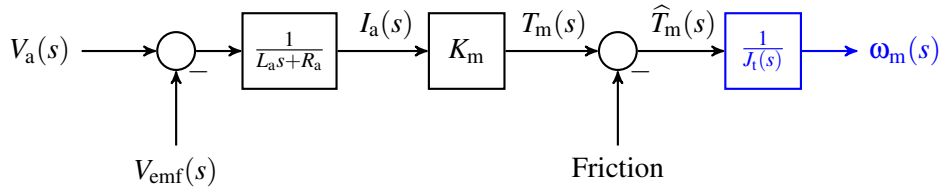
Friction is a fundamental force that resists the motion of an object sliding or rolling over another. The study of friction is in depth and is only touched on here. There are four types of friction

1. Static — occurs between surfaces that are not moving
2. Kinetic — occurs between surfaces that are sliding against one another
3. Rolling — occurs when an object rolls over a surface
4. Fluid — occurs when an object moves through a fluid

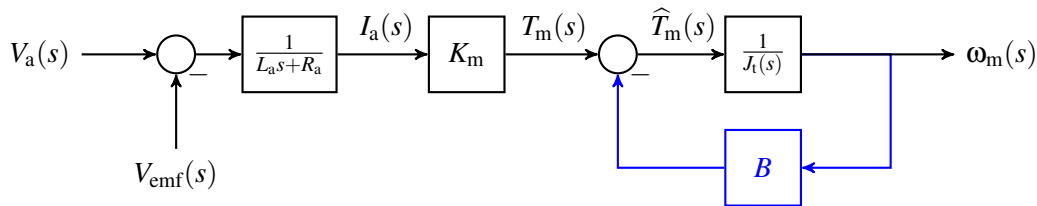
For many controls approaches only kinetic or rolling friction is considered. The frictional coefficient is considered constant, B .



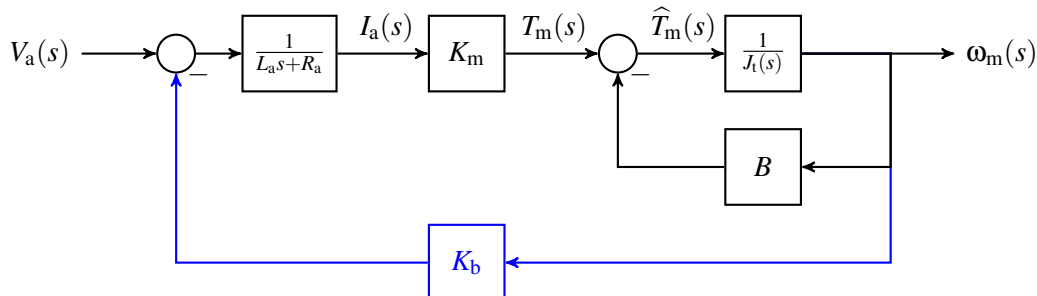
Now, let's add the inertial component to the diagram



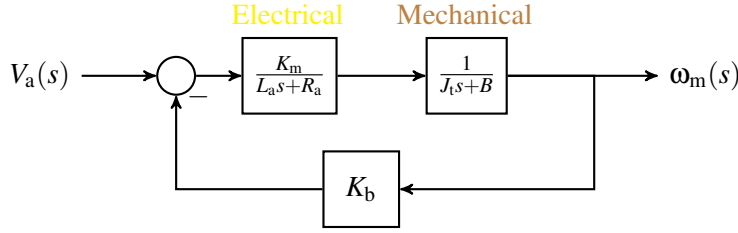
Now, we need to complete the feedback paths. The output is the shaft rotational rate, the rotation produces either kinetic or rolling friction by causing surfaces to slide or roll across one another



Not only does the shaft rotational rate produce the frictional feedback, but it also produces the V_{emf} — function of the physical properties of the motor (e.g. magnet strength, wiring size and proximity to the magnets). The coefficient K_b a measured value and provided by the motor manufacturer.



Simplify the diagram by eliminating the friction feedback and combining the first two transfer functions. This produces a block diagram with the electrical components in one block and the mechanical components in the other



where

1. L_a is the armature inductance
2. R_a is the armature resistance
3. J_t is the total inertia of the system, which is motor inertia plus load inertia, $J_t = J_m + J_L$
4. B is the frictional coefficient
5. K_b is the EMF coefficient
6. $V_a(s)$ is the armature input
7. $\omega_m(s)$ is the shaft rotational rate

Solving the block diagram produces the motor transfer function

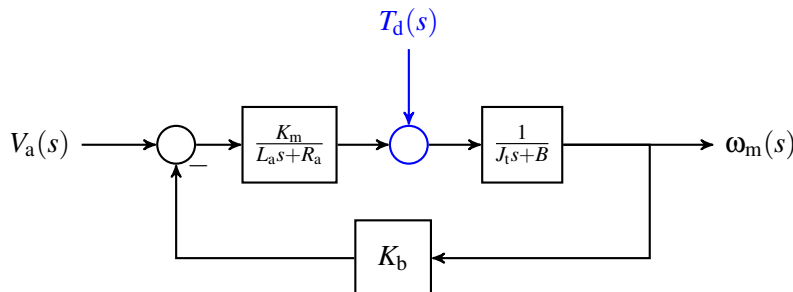
$$\frac{\omega_m(s)}{V_a(s)} = G_m(s) = \frac{K_m}{L_a J_t s^2 + (L_a B + R_a J_t) s + R_a B + K_m K_b} \quad (8.1)$$

Often, the second order model is more precise than is needed in the design, we reduce the complexity of the model by setting $L_a = 0$, which produces a **first order** DC servo model

$$G_m(s) = \frac{K_m}{R_a J_t s + R_a B + K_m K_b} \quad (8.2)$$

8.1 External Disturbances

For practical applications, we often must consider external disturbances. Most devices that move will encounter environmental effects in the form of fluid flow (wind, water) and/or environmental variations (uneven ground, wheel slippage). The best practice is to consider the disturbance as an additional input into the nominal system. Systems with multi-input / multi-output use the principle of super-position to determine the transfer function.



First, the disturbance input is set to zero, $T_d(s) = 0$, and the output with respect to the input (wrt) is the same as above in (8.1). Then, we find $\frac{\omega_m(s)}{T_d(s)}$ by setting $V_a(s) = 0$ and resolving the block diagram.

$$\begin{aligned}
 \frac{\omega_m(s)}{T_d(s)} &= \frac{\frac{1}{J_t s + B}}{1 + \frac{K_m K_b}{(L_a s + R_a)(J_t s + B)}} \\
 &= \frac{\frac{1}{J_t s + B}}{\frac{(L_a s + R_a)(J_t s + B) + K_m K_b}{(L_a s + R_a)(J_t s + B)}} \\
 &= \frac{L_a s + R_a}{(L_a s + R_a)(J_t s + B) + K_m K_b} \\
 \frac{\omega_m(s)}{T_d(s)} = \hat{G}_m(s) &= \frac{L_a s + R_a}{L_a J_t s^2 + (L_a B + R_a J_t)s + R_a B + K_m K_b}
 \end{aligned} \tag{8.3}$$

Then, the output of the disturbed system is

$$\begin{aligned}
 \omega_m(s) &= G_m V_a(s) + \hat{G}_m T_d(s) \\
 \omega_m(s) &= \frac{K_m V_a(s)}{L_a J_t s^2 + (L_a B + R_a J_t)s + R_a B + K_m K_b} + \frac{(L_a s + R_a) T_d(s)}{L_a J_t s^2 + (L_a B + R_a J_t)s + R_a B + K_m K_b}
 \end{aligned} \tag{8.4}$$

Note: the characteristic polynomial is the same for $G_m(s)$ and $\hat{G}_m(s)$.

As above, If we want to reduce the model to a first order, assume the armature inductance is negligible $L_a = 0$

$$\omega_m(s) = \frac{K_m V_a(s) + R_a T_d(s)}{R_a J_t s + R_a B + K_m K_b} \tag{8.5}$$

For either model, **basic** disturbance rejection occurs when $R_a \rightarrow 0$ and $K_m \gg 0$. However, there are times that selecting a motor with ideal parameters is not enough to reject the external disturbance; when that is the case, we must design a disturbance rejecting controller — usually a PID.

Example — Basic Disturbance Rejection

Given: A DC Servo motor system with a large inductance — too big to be ignored; determine the disturbance rejection for two different DC motors.

Motor #1 has the parameters

Parameter	Value
L_a	0.005
R_a	5.3
K_b	0.15
K_m	2.6
B	0.075
J_t	11
B	0.075

which produces a **normalized** output from (8.4) as

$$\omega_m(s) = \frac{47.27V_a(s)}{s^2 + 1060s + 14.32} + \frac{(0.09s + 96.36)T_d(s)}{s^2 + 1060s + 14.32}$$

assuming both the input and the disturbance source are bounded (for the disturbance, think of a constant headwind for a rolling cart) then

$$\begin{aligned}\omega_m(t \rightarrow \infty) &= \frac{47.27}{14.32} \cdot \lim_{s \rightarrow 0} \{sV_a(s)\} + \frac{96.36}{14.32} \cdot \lim_{s \rightarrow 0} \{sT_d(s)\} \\ &= 3.3 \cdot \lim_{s \rightarrow 0} \{sV_a(s)\} + 6.73 \cdot \lim_{s \rightarrow 0} \{sT_d(s)\}\end{aligned}$$

For complete analysis, the two input expressions need to be evaluated (FVT), but even with this incomplete setup, we can see the disturbance input is scaled significantly more than the armature voltage input. This may produce an output overly influenced by the disturbance.

On to **motor #2** which has a lower armature resistance and a larger motor torque gain than the first motor

Parameter	Value
L_a	0.005
R_a	2.3
K_b	0.15
K_m	8.5
J_t	11
B	0.075

then

$$\omega_m(s) = \frac{152V_a(s)}{s^2 + 460s + 26.32} + \frac{(0.09s + 41.82)T_d(s)}{s^2 + 460s + 26.32}$$

assume bounded signals for the inputs

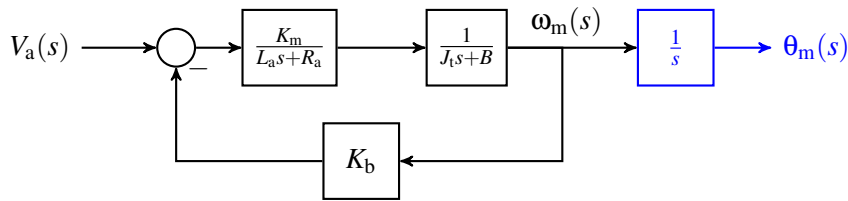
$$\begin{aligned}\omega_m(t \rightarrow \infty) &= \frac{152}{26.32} \cdot \lim_{s \rightarrow 0} \{sV_a(s)\} + \frac{41.82}{26.32} \cdot \lim_{s \rightarrow 0} \{sT_d(s)\} \\ &= 5.78 \cdot \lim_{s \rightarrow 0} \{sV_a(s)\} + 1.58 \cdot \lim_{s \rightarrow 0} \{sT_d(s)\}\end{aligned}$$

When comparing the results to the previous motor, the armature voltage input is upscaled significantly more than the disturbance input. It is likely that the output is more a function of the armature

voltage than the disturbance; however, even with this example the disturbance may contribute more to the output than is desired — or acceptable. For systems that require more disturbance rejection than what is provided by motor selection, we can implement a controller to further suppress the noise.

8.2 Position Tracking

Often the desired output of a DC Servo system is shaft position (degrees or radians) and not shaft rotational rate — especially in robotic and aerospace applications. Mathematically, all one must do is take the output $\omega_m(s)$ and pass it into an integrator. Then, the output of the integrator is the shaft position, $\theta_m(t)$ or $\theta_m(s)$

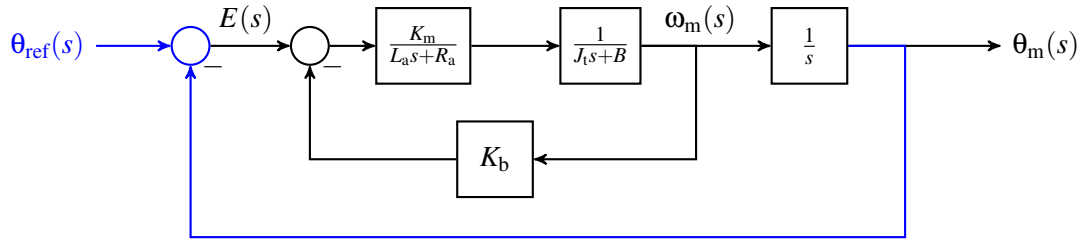


Now, considering the block diagram, the transfer function is

$$\begin{aligned} \frac{\theta_m(s)}{V_a(s)} &= \frac{K_m}{s \left(L_a J_t s^2 + (L_a B + R_a J_t) s + R_a B + K_m K_b \right)} \\ &= \frac{K_m}{L_a J_t s^3 + (L_a B + R_a J_t) s^2 + (R_a B + K_m K_b) s} \end{aligned}$$

We find a system with a pole at the origin (0,0) and two poles in the stable region. Having a pole at the origin produces a marginally stable system that depends in the input signal for BIBO stability.

Normally, we want to direct the output to a given position. Considering a robot arm, we need a joint to move to a set angle and stay there. How do we resolve our desires, verses the model? Simple, apply feedback — negative feedback.



The internal signal $E(s)$ is the difference between the reference input and the actual output. A true error signal that is driving the armature voltage.

The open-loop transfer function of the block diagram is determined to be $\frac{\theta_m(s)}{\theta_{ref}(s)}$ where $\theta_{ref}(s)$ is an input voltage comparable to a shaft angle. The transfer function is found by analyzing the block diagram

$$\frac{\theta_m(s)}{\theta_{\text{ref}}(s)} = \frac{K_m}{L_a J_t s^3 + (L_a B + R_a J_t) s^2 + (R_a B + K_m K_b) s + K_m} \quad (8.6)$$

Now, with (8.6) we have a complete third order model where the input is a *reference* angle and the output is a shaft angle, $\theta_m(s)$. Just as above, we may reduce the order of the model by assuming the armature inductance is zero, $L_a = 0$. When this is the case, the model reduces to a second order representation

$$\frac{\theta_m(s)}{\theta_{\text{ref}}(s)} = \frac{K_m}{R_a J_t s^2 + (R_a B + K_m K_b) s + K_m} \quad (8.7)$$

FVT for the position tracking system, regardless of the model order, when the input is a setpoint

$$\theta_m(t \rightarrow \infty) = \lim_{s \rightarrow 0} s \theta_m(s) = \frac{K_m}{K_m} \rightarrow 1$$

Thus, for a setpoint (unit step) input the output matches the input. The time it takes for the shaft to reach and settle to the final value is determined by the time constants (poles) of the characteristic polynomial. Also, before trusting the FVT we need to verify BIBO stability.

Note: from a practical perspective, we do not use an integrator in the system. The shaft position is achieved by attaching an encoder to the shaft that provides the position. The math model requires an integrator to correctly represent the motor with a position encoder, but we do not need to design an integrator to achieve shaft position output.

Example — Position Tracking

Take the general transfer function (8.6); then using MATLAB apply a setpoint input and observe the output when

Parameter	Value
L_a	0.0001
R_a	1.3
K_b	0.25
K_m	4
J_t	1
B	0.25

Then, the transfer function is

$$\frac{\theta_m(s)}{\theta_{\text{ref}}(s)} = \frac{4}{0.0001s^3 + 1.3s^2 + 1.325s + 4}$$

The poles to the characteristic equation are $-13000, -0.51 \pm 1.68j$ which confirms the system is stable with a complex pair of poles and a real pole. Note the two complex poles are near the $j\omega$ -axis and the real pole is very far to the left.

The two complex conjugate pair poles with real parts at -0.51 are the **dominate** poles in the system and the pole at -13000 is a **transient** pole that only contributes to the output for about five time constants. Dominate poles are the primary driver in the signal response to an input; while the transient poles only contribute the the output for a very short period of timer early in the response.

Let's consider the time constant of the real pole

$$\tau_r = \frac{1}{13000} = 0.000077$$

while the two complex poles have a time constant

$$\tau_c = \frac{1}{0.51} = 1.96$$

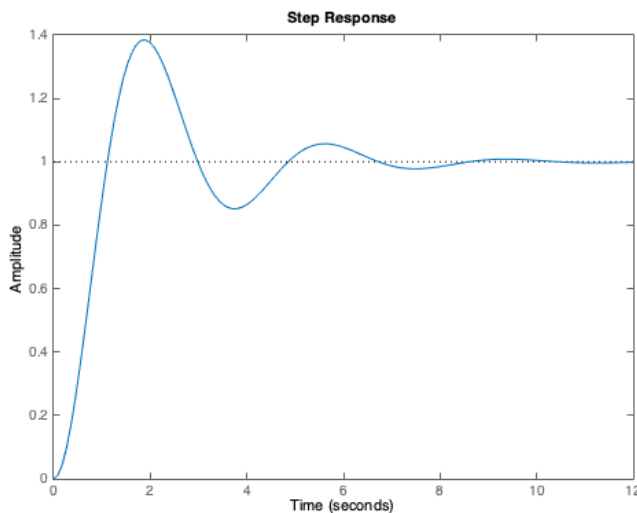
The **rule of thumb for dynamic response to a pole**: the dynamic contribution to the signal only lasts for about five time constants of a stable expression. This means the real pole far to the left is a transient signal and contributes to the dynamic output response for

$$5 \cdot \tau_r = 0.00038\text{sec}$$

while the dominant poles near the origin contribute to the dynamic output for

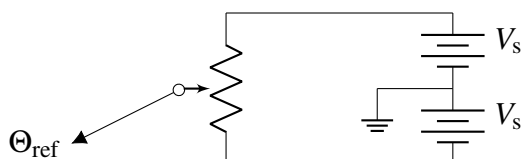
$$5 \cdot \tau_c = 9.8\text{sec}$$

So, the contribution from the transient pole is inconsequential to the dynamic response of the system, as seen here



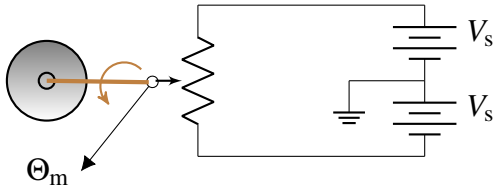
8.2.1 Rotary Position Tracking System

We use two power supplies connected in tandem and a potentiometer to produce the reference voltage



The user rotates the rheostat to produce a voltage between $+V_s$ and $-V_s$ by tuning the reference to the desired voltage.

Then, the position of the motor shaft is determined in a similar manner, but the potentiometer is directly connected to the motor shaft so that when the shaft rotates, the wiper on the potentiometer moves along the coils of the rheostat producing a voltage for the shaft position.

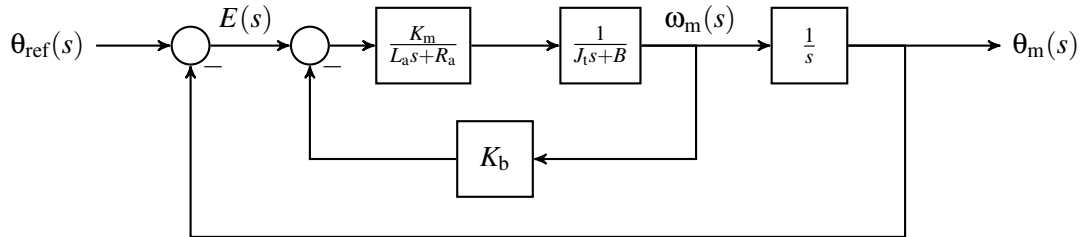


The potentiometer at the motor determines the range of rotation. A single-turn pot that has a 270° total angle of deflection will limit the movement of the shaft to 135° for $+V_s$ and -135° for $-V_s$. The motion between the limits will be linear with 0° at zero volts.

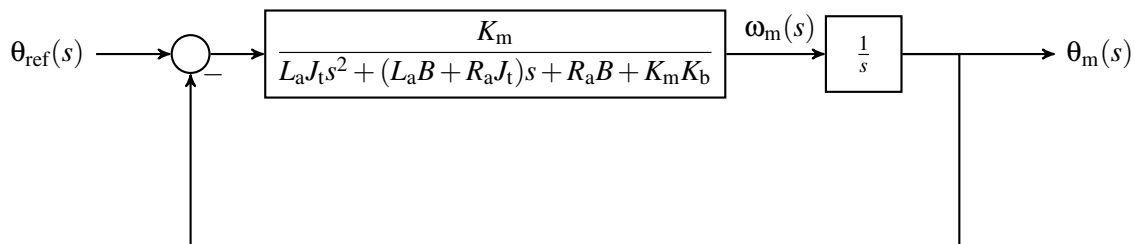
8.2.2 Tunable Gains with Tachometer Feedback

Many motors with embedded controllers have readily available tunable gain parameters — a proportional gain in the forward path with $K > 0$ and an internal feedback gain parameter called tachometer feedback $K_f \geq 0$.

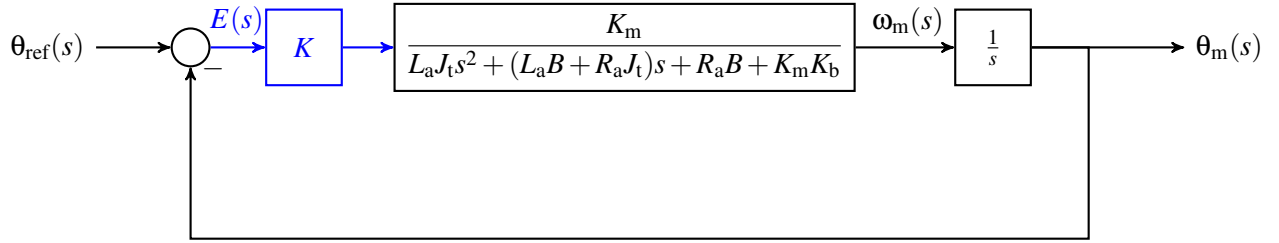
Let's begin with the expanded model



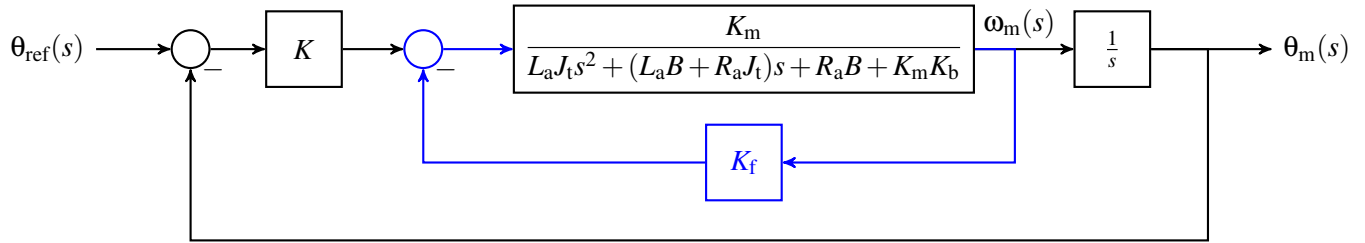
now, simplify the inner feedback loop



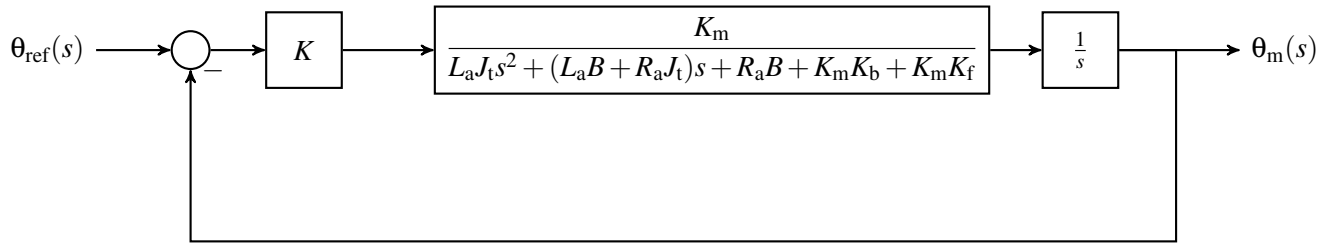
The proportional gain is between the summation junction and the first motor block so that the gain K modifies the error signal.



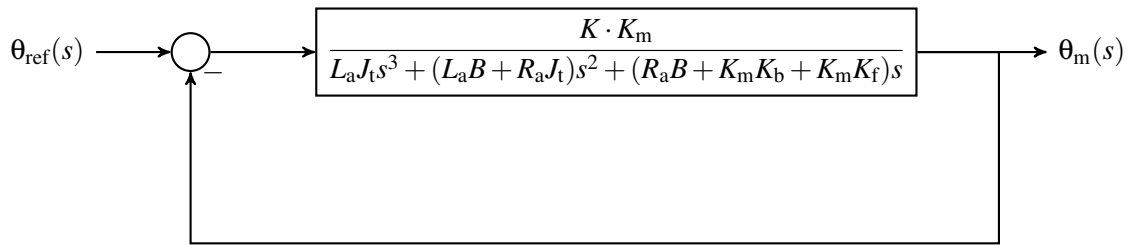
The tachometer feedback is about the motor transfer function



Now that we have a complete block diagram, let's simplify to a transfer function



Combine the three cascaded blocks in the forward path



and finally eliminate the remaining feedback loop we have a transfer function with the two tunable parameters — the proportional gain K and the tachometer feedback K_f .

$$\frac{\theta_m(s)}{\theta_{ref}(s)} = \frac{K \cdot K_m}{L_a J_t s^3 + (L_a B + R_a J_t)s^2 + (R_a B + K_m K_b + K_m K_f)s + K \cdot K_m} \quad (8.8)$$

Additionally, we can develop a simpler model by setting the armature inductance to zero

$$\frac{\theta_m(s)}{\theta_{ref}(s)} = \frac{K \cdot K_m}{R_a J_t s^2 + (R_a B + K_m K_b + K_m K_f)s + K \cdot K_m} \quad (8.9)$$

From here, we can use all of the other techniques taught in class to analyze the system and tune the response with the variables at our disposal.

Example — Basic Position Tracking with K & K_f

Given the same motor from a previous example

Parameter	Value
L_a	0.005
R_a	2.3
K_b	0.15
K_m	8.5
J_t	11
B	0.075

we find the third order transfer function

$$\frac{\theta_m(s)}{\theta_{ref}(s)} = \frac{8.5K}{0.06s^3 + 25.3s^2 + (1.45 + 8.5K_f)s + 8.5K}$$

Now, let us take a look at the poles for different tunings

K	K_f	Poles
1	0	$-459.95, -0.03 \pm 0.58j$
10	0	$-459.96, -0.03 \pm 1.83j$
10	1	$-459.62, -0.19 \pm 1.82j$
10	5	$-458.27, -0.87 \pm 1.62j$
10	10	$-456.57, -1.72 \pm 1.66j$
3	5	$-458.27, -0.87 \pm 0.50j$

One can look at the poles for the various tunings and empirically choose $K = 3$ and $K_f = 5$ as the best possible tuning on the list. Simply because the dominate poles (the ones near the vertical axis) have a greater ratio of real to imaginary magnitudes. This means the output oscillations are less than the other choices. Notice the transient pole is mostly immune to changes in the tunings and is far from the vertical axis. This is the pole that is eliminated when reducing the order of the model by setting the armature inductance to zero. Let's take a brief look at the poles for the reduced order model

K	K_f	Poles
10	1	$-0.19 \pm 1.82j$
10	5	$-0.87 \pm 1.61j$
3	5	$-0.87 \pm 0.50j$

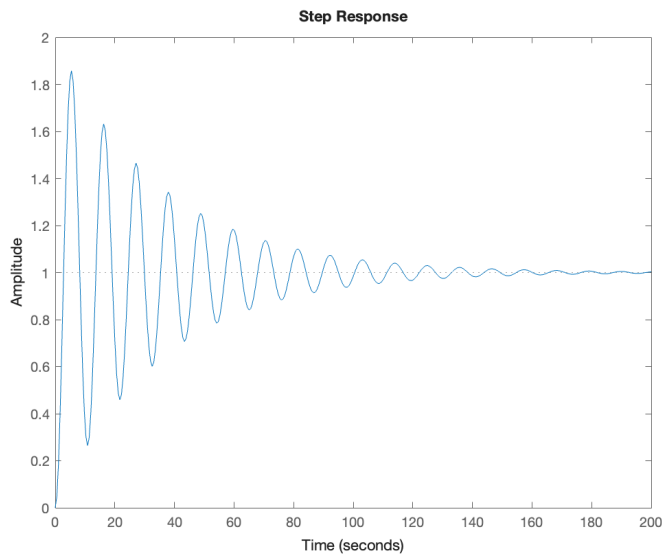
Rounding to the hundredths place, the results for the dominant poles are the same.

There are times and systems where precision is needed and the full-model is required. Other times, the reduced order model is sufficient.

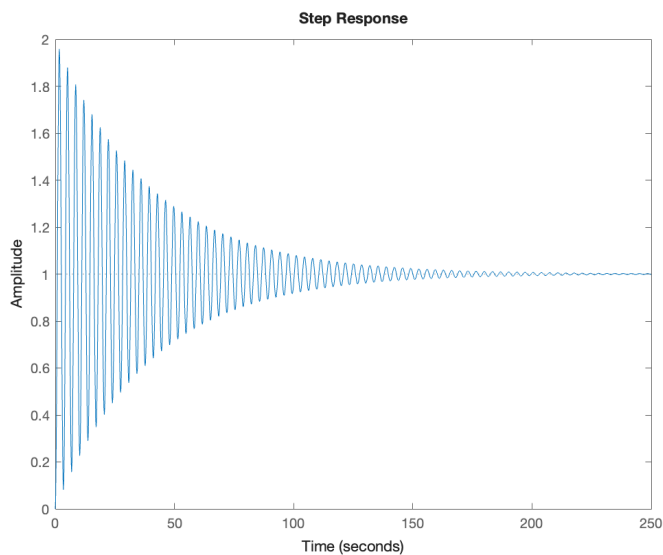
Let's take a look at the full-order model output response to a step input using MATLAB and compare to the poles. How will the selection of $K = 3$, K_f hold up to the modeling results?

Output response to a setpoint input for the various tunings

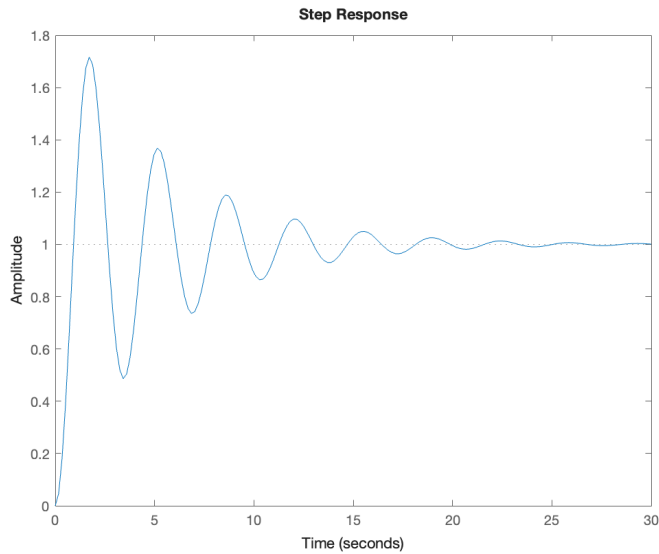
Set the parameters to $K = 1$, $K_f = 0$ and run the model. We see significant output oscillations and a settling time of about 180 seconds.



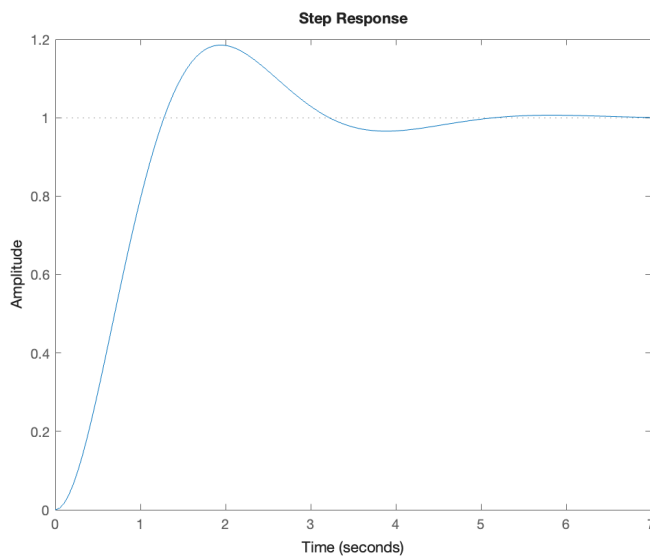
Now, set the parameters to $K = 10$, $K_f = 0$ and repeat the run. Now, the oscillations are greater in magnitude and frequency than before and the sell looks to have increased slightly — about 200 seconds.



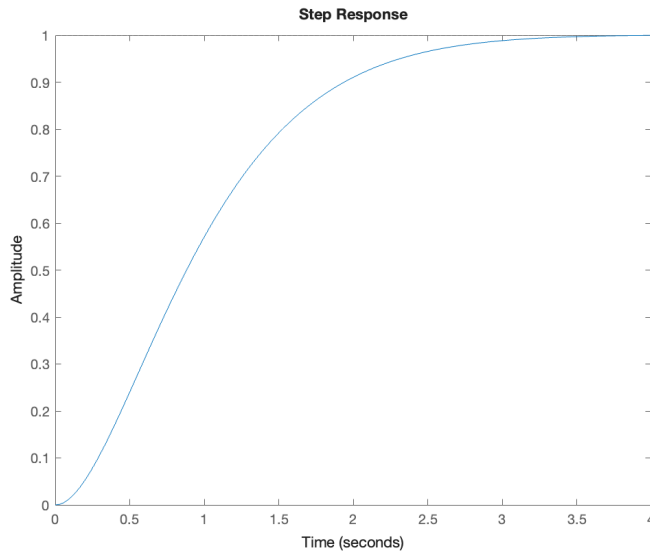
$K = 10$, $K_f = 1$ and repeat the run. Now, with the addition of tachometer feedback, the output oscillations are significantly reduced in frequency and the settling time is greatly reduced — to about 25 seconds.



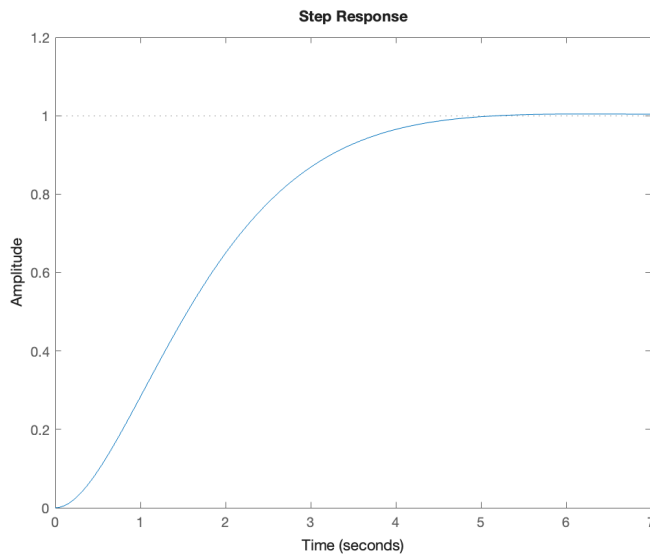
$K = 10$, $K_f = 5$. Now, the oscillations are still present, but there are only two peaks before steady state at about six seconds.



$K = 10$, $K_f = 10$ The result here appears to have no oscillations and goes to steady state in about 3.5 seconds.



$K = 3$, $K_f = 5$. The final result produces an output without oscillation and a settling time of about five seconds.



When considering the output response to a setpoint input only the last two tunings produce an output response with minimal oscillations. Setting $K = K_f = 10$ produces the shortest settling time; about four seconds. While $K = 3$ and $K_f = 5$ produces a settling time slightly longer; about four and a half seconds. Either produce a nice shaped response at the output. Generally, selecting the results with the smaller gain is preferable just to minimize the energy necessary to produce the desired result — think of minimizing the armature current when possible.